

Title of the Course: React and Node JS Lab					Semester: 5th	
Course Code:			BRJSB507		CIE Marks	50
Course Type:			React and Node JS Lab		SEE Marks	50
Student Learning Hours (L:T:P:S)			0:0:28:14		Total Marks	100
Credits			1		Exam Hours	03
Examination Type (SEE)			Laboratory			
L - Lecture	T – Tutorial	P – Practical	S – Self Study	CL - Cognitive Level	PL - Psychomotor Level	
COs	Course Outcome: At the end of the course, the students will be able to				CL (Highest Level)	PL (Highest Level)
CO1	Develop full-stack web applications using React components, state management, Node.js, REST APIs, and MongoDB connectivity.				CL3	PL2\3
CO2	Implement user authentication, user registration, login, logout, profile management, password handling, and input validation in web applications.				CL3	PL3
CO3	Perform CRUD operations, database design, data storage, retrieval, search, filtering, sorting, and data management using MongoDB.				CL3	PL3
CO4	Design and develop responsive user interfaces, component-based applications, product management systems, search functionality, filtering mechanisms, and shopping cart operations using React.				CL4	PL4
CO5	Integrate external APIs, order processing, state management, real-time data handling, backend services, and end-to-end e-commerce workflows in full-stack applications.				CL4	PL4
CO – Assessment Mapping						
Course Outcomes	Continuous Internal Assessment				Semester End Exam	
	CIE I	CIE II	Assignment/Activities			
	15%	15%	20%		50%	
CO 1	√		√		√	
CO 2	√		√		√	
CO 3	√	√	√		√	
CO 4		√	√		√	
CO5		√			√	
<u>PRACTICAL COMPONENTS</u>						
1.	Develop a User Authentication System using React, Node.js, and MongoDB. Requirements: - Create a user registration form. - Implement user login functionality.					

	• Validate email and password fields. • Store user details in MongoDB. • Implement logout functionality
2.	Design and develop a User Profile Management System using React, Node.js, and MongoDB. Requirements: • User Registration • User Login • View Profile • Edit Profile • Change Password • Input Validation • Store and retrieve user data from MongoDB
3.	Develop a Product Management System with search and filtering features using React, Node.js, and MongoDB. Requirements: • Display products in a grid layout • Add products to MongoDB • Search products by name • Filter products by category • Sort products by price • View product details • Responsive user interface
4.	Build a Shopping Cart System using React, Node.js, Express.js, and MongoDB. Requirements: • Add products to cart • Remove products from cart • Update product quantity • Display cart summary • Calculate total amount • Store cart information in MongoDB • Maintain user-specific cart data
5.	Develop an Order Processing System using React, Node.js, and MongoDB. Requirements: • Capture customer details • Manage delivery address • Display order summary • Place orders • Store order details in MongoDB • Generate order confirmation • View order history • Track order status

Part- B

Capstone Project

Full-Stack E-Commerce Web Application

Modules to Include

1. User Authentication

- User Registration
- Login
- Logout
- JWT Authentication
- Password Hashing (bcrypt)
- Input Validation

2. User Profile Management

- View Profile
- Edit Profile
- Change Password
- Upload Profile Picture

3. Product Management

- Add Product
- Update Product
- Delete Product
- View Products
- Product Categories
- Product Images

4. Search, Filter & Sort

- Search by Product Name
- Filter by Category
- Sort by Price (Low → High, High → Low)

5. Shopping Cart

- Add to Cart
- Remove from Cart
- Update Quantity
- Cart Total Calculation
- User-Specific Cart Storage

6. Order Processing

- Checkout Page
- Customer Details
- Delivery Address
- Order Summary
- Place Order
- Order History
- Order Tracking Status

7. Admin Panel

- Admin Login
- Manage Users
- Manage Products
- View All Orders
- Update Order Status